Chapter 9: The Behavior of Solutions

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9.1. Introduction

✓ Ideal gases

- No interaction
- Volumeless
- Have the same thermodynamic mixing properties
- Its configurational entropy has its highest possible value
- Interaction lowers the entropy
- Decreasing temperature (T) → gas condensing to liquid or solid

✓ Condensed phases:

- Have interactions (determined by atomic size, electronegativity, and electron-to-atom ratio);

✓ Solution thermodynamics

- Relationships of the vapor pressure–T–composition

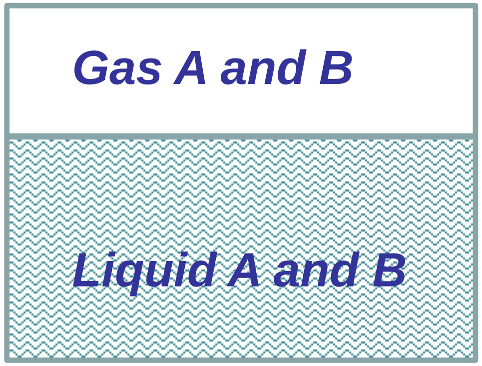
9.2. Raoult's Law and Henry's Law

✓ *Raoult's law*

- The vapor pressure (p_i) exerted by a component *i* in a solution is equal to the product of the mole fraction of *i* (X_i) in the solution and the saturated vapor pressure of pure liquid *i* (p_i^o) at the temperature of the solution.

$$p_A = X_A p_A^0$$

$$p_B = X_B p_B^0$$



A binary solution A--B

9.2. Raoult's Law and Henry's Law

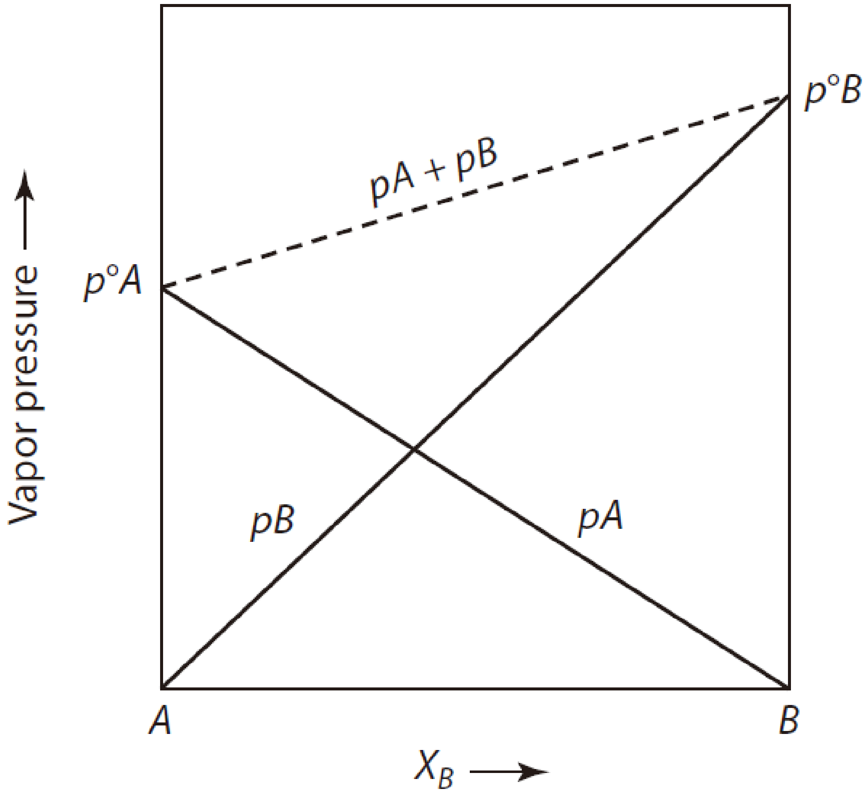


Figure 9.1 The vapor pressures exerted by the components of a binary solution as a function of composition. Both components of this solution obey Raoult's law at all compositions.

9.2. Raoult's Law and Henry's Law

- ✓ If a quantity of **pure liquid *A*** is placed in a closed, initially evacuated vessel at the temperature T, some of the liquid will spontaneously evaporate until the pressure in the vessel reaches the saturated vapor pressure of liquid A, P_A^o, at the temperature T.
- ✓ In this state, a dynamic equilibrium is established in which the rate of evaporation of liquid A equals the rate of condensation of the vapor A.

9.2. Raoult's Law and Henry's Law

- ✓ The rate of evaporation, $r_{e(A)}$, is determined by the magnitude of the energy of the bonds between the atoms of *A* at the surface of the liquid.

- ✓ The forces exerted between the atoms are such that each surface atom is located near the bottom of a potential energy well, and for an atom to leave the surface of the liquid and enter the vapor phase, it must acquire an activation energy, E^*.

- ✓ The intrinsic rate of evaporation, $r_{e(A)}$, is determined by the depth of the potential energy well—that is, by the magnitude of E^*—and by the temperature T.

9.2. Raoult's Law and Henry's Law

- ✓ The rate of condensation, $r_{c(A)}$, is proportional to the number of *A* atoms in the vapor phase which strike the surface of the liquid in a unit time.
- ✓ For a given temperature, this is proportional to the pressure of the vapor. Thus, $r_{c(A)} = k^A P_A^o$, and **at equilibrium,** $r_{e(A)} = r_{c(A)}$, and thus,

$$r_{e(A)} = k^A p_A^\circ$$

9.2. Raoult's Law and Henry's Law

- ✓ The energies of the atoms at the surface are quantized, and the distribution of the surface atoms among the available quantized energy levels is given as

$$n_i = \frac{n \exp\left(-E_i / k_B T\right)}{Z}$$

where n_i/n is the fraction of the atoms in the E_i^{th} energy level, and Z, the partition function, is given by

$$Z = \sum_{0}^{\infty} \exp\left[-\frac{E_i}{k_B T}\right]$$

9.2. Raoult's Law and Henry's Law

- ✓ If the quantized energy levels are spaced closely enough that the summation can be replaced by an integral, then

$$Z = \int_0^\infty \exp\left[-\frac{E_i}{k_B T}\right] dE = kT$$

which is the average energy per atom.

- ✓ The fraction of surface atoms which have energies greater than the activation energy for evaporation, *E**, is

$$\frac{n_i^*}{n} = \frac{1}{k_B T}\int_{E^*}^\infty \exp\left[-\frac{E_i}{k_B T}\right] dE = \exp\left[-\frac{E^*}{k_B T}\right]$$

9.2. Raoult's Law and Henry's Law

$$r_{e(A)} = k^{A} p_{A}^{\circ}$$

$$\frac{n_i^*}{n} = \frac{1}{k_B T}\int_{E^*}^{\infty} \exp\left[-\frac{E_i}{k_B T}\right] dE = \exp\left[-\frac{E^*}{k_B T}\right]$$

- ✓ The evaporation rate, $r_{e(A)}$, is proportional to n_i^*/n , and thus, it increases exponentially with increasing temperature and decreases exponentially with the increasing value of E^*.
- ✓ This illustrates why the saturated vapor pressures of liquids are exponential functions of temperature.

9.2. Raoult's Law and Henry's Law

- ✓ Consider the effect of the addition of a small quantity of liquid B to liquid A . If the mole fraction of A in the solution is X_A and the atomic diameters of A and B are similar, then, assuming that the composition of the surface of the liquid is the same as that of the bulk liquid, the fraction of the surface sites occupied by A atoms is X_A .

- ✓ Since *A* can only evaporate from surface sites occupied by *A* atoms, the rate of evaporation of *A* is **decreased by the factor** X_A, and since the rates of evaporation and condensation are equal to one another at equilibrium, the equilibrium vapor pressure of *A* exerted by the *A–B* solution is decreased from P_A^o to P_A.

9.2. Raoult's Law and Henry's Law

$$r_{e(A)} X_A = k^A p_A$$

$$r_{e(A)} = k^A p_A^{\circ}$$

$$p_A = X_A p_A^{\circ}$$

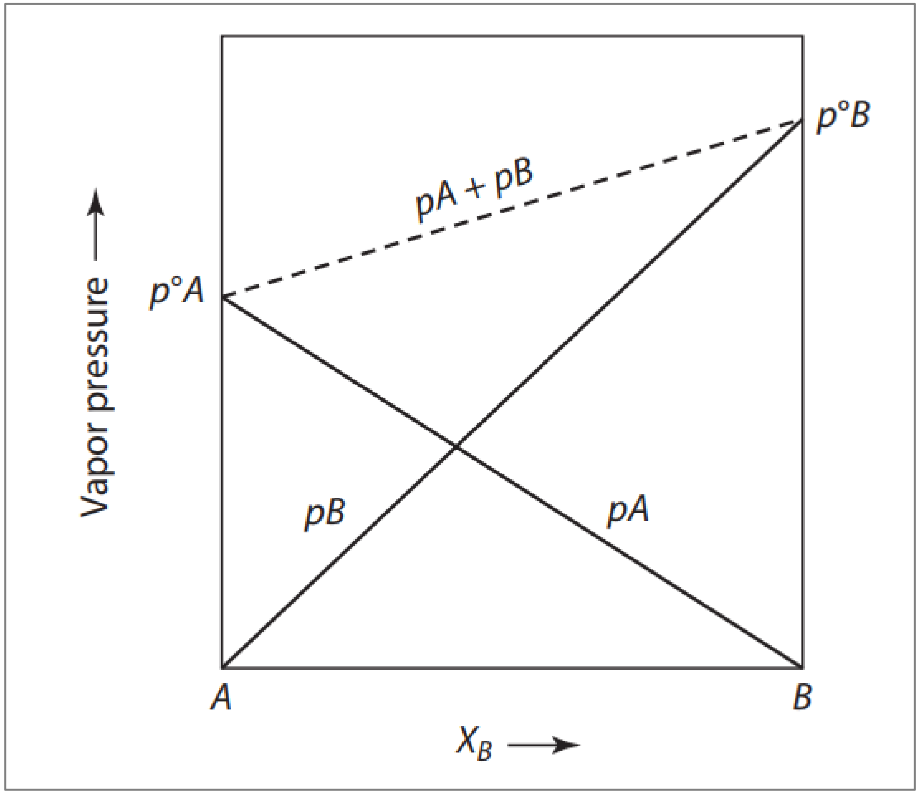


Figure 9.1

The vapor pressures exerted by the components of a binary solution as a function of composition. Both components of this solution obey Raoult' s law at all compositions.

9.2. Raoult's Law and Henry's Law

✓ When **pure liquid *B*** is placed in an initially evacuated vessel at the temperature *T*, equilibrium between the liquid and its vapor phases occurs when

$$r_{e(B)} = k^{B} p_{B}^{\circ}$$

For liquid *B* containing **a small amount** of *A*,

$$r_{e(B)} X_{B} = k^{B} p_{B}$$

$$\boxed{\begin{aligned} r_{e(B)} &= k^{B} p_{B}^{\circ} \\ r_{e(B)} X_{B} &= k^{B} p_{B} \end{aligned}} \quad \Rightarrow \quad p_{B} = X_{B} p_{B}^{\circ}$$

9.2. Raoult's Law and Henry's Law

$$p_A = X_A p_A^{\circ} \qquad p_B = X_B p_B^{\circ}$$

- ✓ They are expressions of ***Raoult's law***, which states that the vapor pressure exerted by a component *i* in a solution is equal to the product of the mole fraction of *i* in the solution and the saturated vapor pressure of pure liquid *i* at the temperature of the solution.
- ✓ The components of a solution which obeys Raoult's law are said to exhibit Raoultian behavior.

9.2. Raoult's Law and Henry's Law

$$r_{e(A)} X_A = k^A p_A \qquad r_{e(B)} X_B = k^B p_B$$

- ✓ The derivations these equations require the **assumption** that the intrinsic rates of evaporation of *A* and *B* are independent of the composition of the solution.

$$E_{A\text{-}A} = E_{B\text{-}B} = E_{A\text{-}B}$$

- ✓ This requires that the magnitudes of the *A–A*, *B–B*, and *A–B* bond energies in the solution be identical, in which case the depth of the potential energy well of an atom at the surface is independent of the types of atoms which it has as its nearest neighbors.

- ✓ When this happens, the arrangement of the atoms does not depend on the bond energies, and hence, they will take a configuration which maximizes the configurational entropy of the solution.

9.2. Raoult's Law and Henry's Law

- ✓ Consider the case in which the ***A–B*** bond energy is considerably **more negative** than the ***A–A*** and ***B–B*** bond energies,

$$E_{A-B} < E_{B-B} \qquad E_{A-B} < E_{A-A}$$

Consider a solution of *B* in *A* which is **sufficiently dilute** that every *B* atom on the surface of the liquid is surrounded only by *A* atoms.

- ✓ In this case, the *B* atoms at the surface are each located in a **deeper** potential energy well than are the *B* atoms at the surface of pure *B*. Thus, in order to leave the surface and enter the vapor phase, the *B* atoms have to **overcome larger** energy barriers,

9.2. Raoult's Law and Henry's Law

✓ the intrinsic rate of evaporation of *B* is decreased from $r_{e(B)}$ to $r'_{e(B)}$. Equilibrium between the condensed solution and the vapor phase occurs when

$$r'_{e(B)} X_B = k^B p_B$$

$$r_{e(B)} = k^B p_B^\circ$$

$$p_B = \frac{r'_{e(B)}}{r_{e(B)}} X_B p_B^\circ$$

$r'_{e(B)} < r_{e(B)}$

$$p_B = k^B X_B$$

$k^B < 1$

9.2. Raoult's Law and Henry's Law

- ✓ As the mole fraction of *B* in the *A*–*B* solution is increased, the probability that all of the *B* atoms on the surface of the liquid are surrounded only by *A* atoms decreases.

- ✓ The occurrence of a pair of neighboring *B* atoms on the surface decreases the depth of the potential wells in which they are located and, hence, increases the value of $r'_{e(B)}$.

9.2. Raoult's Law and Henry's Law

$$p_B = k^B X_B$$

- Beyond some critical mole fraction of A, $r'_{e(B)}$ varies with composition, and hence, the equation is **no longer obeyed by *B*** in solution.

- The equation is obeyed only over an initial range of concentration of *B* in *A*, the extent of which is dependent on the temperature of the solution and on the relative magnitudes of the *A–A*, *B–B*, and *A–B* bond energies.

9.2. Raoult's Law and Henry's Law

- ✓ *A* similar consideration of dilute solutions of *A* in *B* gives

$$p_A = k^A X_A$$

which is obeyed over an **initial range of concentration**.

- ✓ *Henry's Law*,

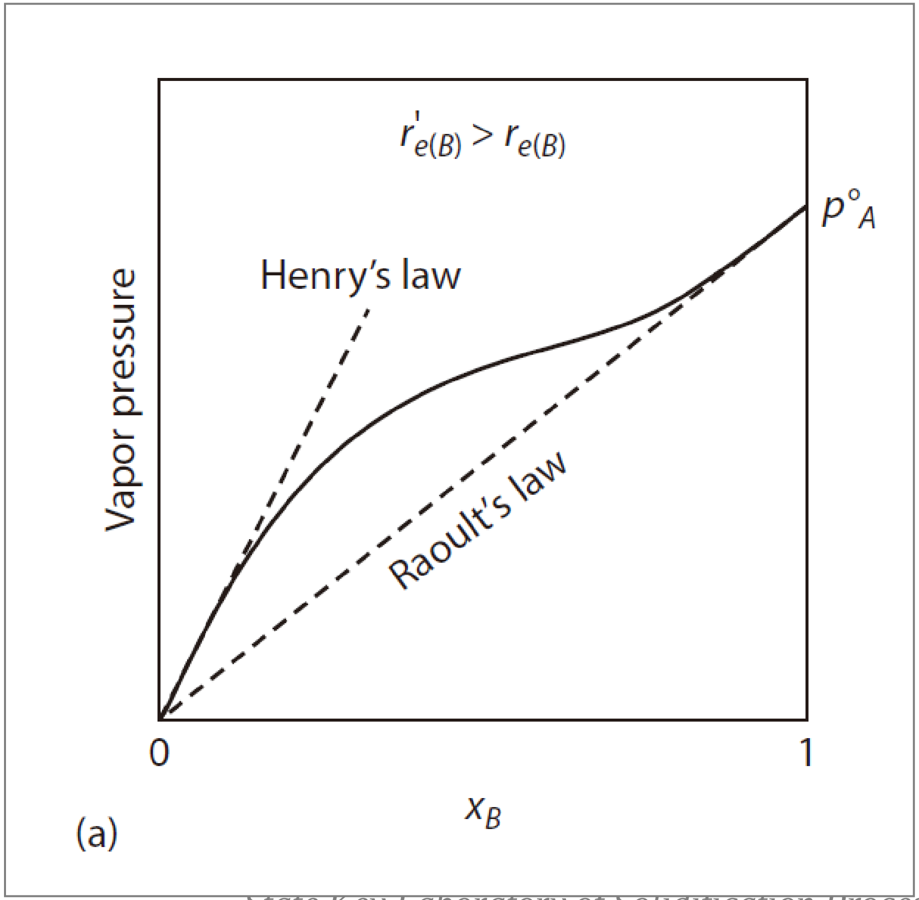
$$p_A = k^A X_A \qquad\qquad p_B = k^B X_B$$

- In the ranges of composition in which Henry's law is obeyed, the **solutes** are said to exhibit *Henrian* behavior.

Raoult's law $\quad p_A = X_A p_A^\circ \qquad p_B = X_B p_B^\circ$

9.2. Raoult's Law and Henry's Law

✓ If the *A–B* bond energy is less negative than the *A–A* and *B–B* bond energies, then, since $r'_{e(B)} > r_{e(B)}$, the Henry's law line lies above the *Raoult's law* line.

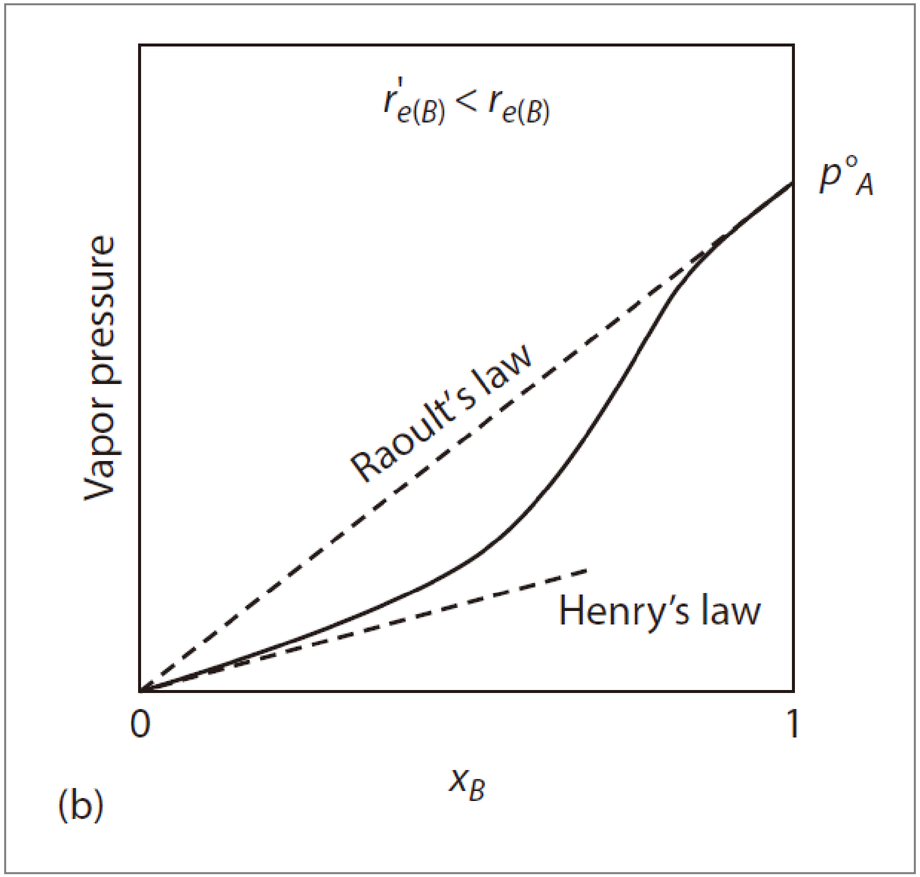


$$p_B = k^B X_B$$

Figure 9.2 (a) The vapor pressure of a solute of a binary solution which exhibits positive deviation from Raoultian behavior.

9.2. Raoult's Law and Henry's Law

✓ If the *A–B* bond energy is more negative than the *A–B* and *B–B* bond energies. In this case, $r'_{e(B)} < r_{e(B)}$, and hence, the Henry's law line for the solute lies below the Raoult's law line.



$$p_B = k^B X_B$$

Figure 9.2
(b) The vapor pressure of a solute of a binary solution which exhibits negative deviation from Raoultian behavior.

Outlines

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9.5 The Gibbs Free Energy of Formation of a Solution

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9.3. The Thermodynamic Activity of a Component in Solution

✓ The thermodynamic activity of a component in any state at the temperature *T* is formally defined as being the **ratio of the fugacity** of the substance **in that state to its fugacity** in its standard state.

✓ For the species or substance *i*,

$$\text{Activity of } \ i \equiv a_i \equiv \frac{f_i}{f_i^\circ}$$

In a condensed solution, f_i is the fugacity of the component *i* **in the solution** at the temperature *T*, and f_i^o is the fugacity of **pure** *i* (the standard state) at the temperature *T*.

9.3. The Thermodynamic Activity of a Component in Solution

- ✓ If the vapor in **equilibrium** with the condensed solution is **ideal**, then $f_i = p_i$, in which case

$$a_i = \frac{p_i}{p_i^{\circ}}$$

That is, the **activity of** ***i*** in a solution, with respect to pure ***i***, **is the ratio** of the partial pressure of ***i*** exerted by the solution to the saturated vapor pressure of pure ***i*** at the same temperature.

$$p_A = X_A p_A^{\circ}$$

- ✓ If the component ***i*** exhibits Raoultian behavior, then

$$p_B = X_B p_B^{\circ}$$

$$a_i = X_i$$

This is an alternative expression of Raoult's law

9.3. The Thermodynamic Activity of a Component in Solution

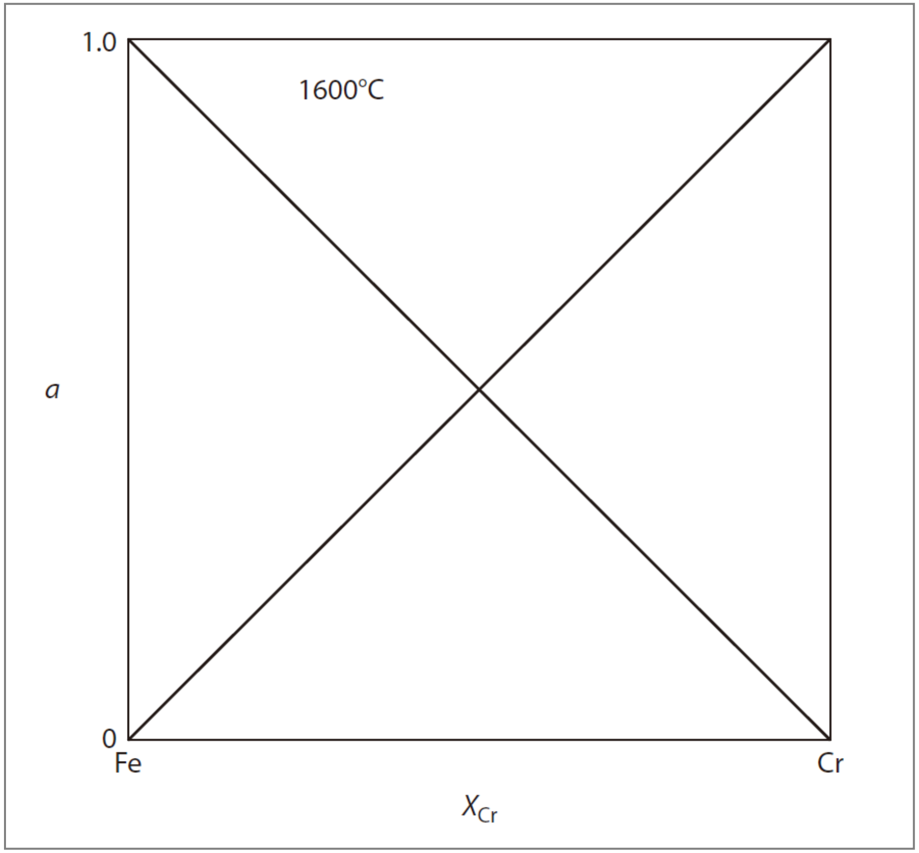


Figure 9.3. The Raoultian behavior over all compositions in a liquid binary solution of Fe and Cr, in terms of the **activities** of the two components.

- ✓ The definition of **activity** normalizes the **vapor pressure–composition relationship** with respect to the **saturated vapor pressure** exerted in the standard state.
- ✓ In an equilibrium state, the activity can never exceed unity.

9.3. The Thermodynamic Activity of a Component in Solution

✓ Over the composition range in which Henry's law is obeyed by the solute *i*, we have

$$a_i = \frac{p_i}{p_i^\circ}$$

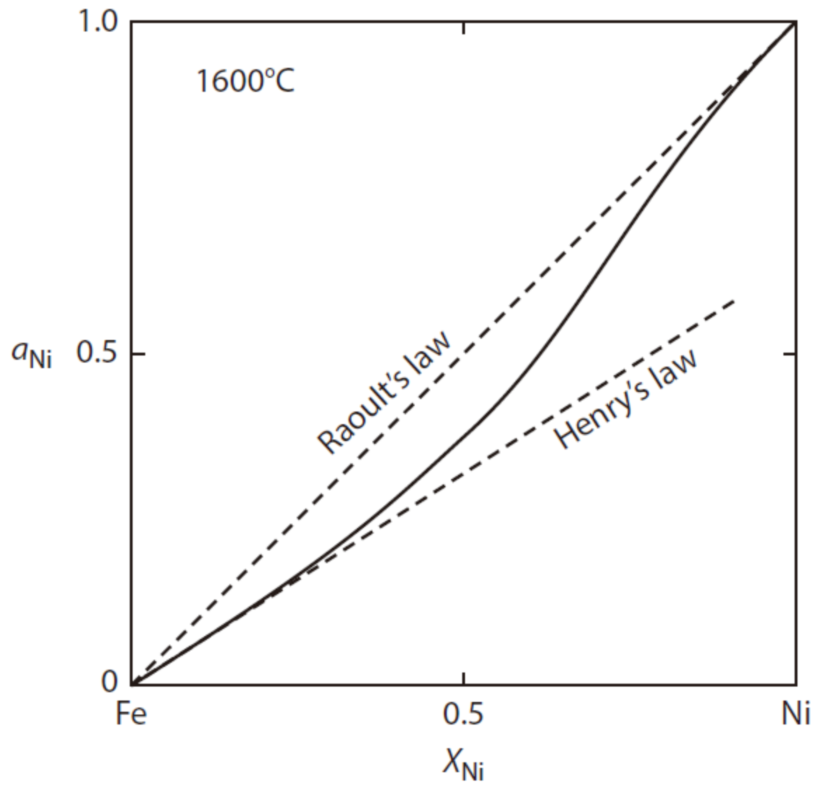
$$p_B = k^B X_B$$

$$a_i = \frac{k^i X_i}{p_i^\circ} = k_{(i)} X_i$$

This is an alternative expression of Henry's law

9.3. The Thermodynamic Activity of a Component in Solution

- ✓ It can be seen that at $X_{Ni} = 0.5$, the Ni atoms in a liquid solution of Fe–Ni are less "active" than they would be if such a solution were a Raoultian one.



- This implies that the Ni atoms are bonded more tightly to Fe atoms than they are to other Ni atoms; that is, $E_{Ni–Fe} < E_{Ni–Ni}$.
- The liquid Fe–Ni system exhibits **negative deviation** from ideality.

Figure 9.4 The **activity** of nickel in the liquid **iron– nickel** system at 1600 °C displaying negative deviation from Raoultian behavior.

Outlines

9.4. The Gibbs-Duhem Equation

- ✓ It is frequently found that the **extensive** thermodynamic **variables** of **only one component** of a binary (or multicomponent) solution are amenable to experimental measurement.
- ✓ In such cases, the corresponding extensive variables of **the other component** can be obtained from a general relationship between the values of the properties of both components.
- ✓ This relationship, which is known as the **Gibbs–Duhem relationship**, is introduced

9.4. The Gibbs-Duhem Equation

- ✓ The value of an **extensive** thermodynamic variable (**state function**) of a solution is a function of the **temperature**, the **pressure**, and the **numbers of moles** of the components of the solution; that is,

- ✓ if ***Q*** is an extensive molar property then

$$Q' = Q'\left(T, P, n_i, n_j, n_k, \cdots\right)$$

- ✓ At constant *T* and *P*, the variation of *Q′* with the composition of the solution is given as

$$dQ' = \left[\frac{\partial Q'}{\partial n_i}\right]_{T,P,n_j,n_k,\ldots} dn_i + \left[\frac{\partial Q'}{\partial n_j}\right]_{T,P,n_i,n_k,\ldots} dn_j + \left[\frac{\partial Q'}{\partial n_k}\right]_{T,P,n_i,n_j,\ldots} dn_k + \cdots$$

9.4. The Gibbs-Duhem Equation

$$dQ' = \left[\frac{\partial Q'}{\partial n_i}\right]_{T,P,n_j,n_k,\ldots} dn_i + \left[\frac{\partial Q'}{\partial n_j}\right]_{T,P,n_i,n_k,\ldots} dn_j + \left[\frac{\partial Q'}{\partial n_k}\right]_{T,P,n_i,n_j,\ldots} dn_k + \cdots$$

The **partial molar** value of an extensive property of a component

$$\overline{Q}' = \left[\frac{\partial Q'}{\partial n_i}\right]_{T,P,n_j,n_k,\ldots}$$

$$dQ' = \overline{Q_i} dn_i + \overline{Q_j} dn_j + \overline{Q_k} dn_k + \cdots$$

9.4. The Gibbs-Duhem Equation

$$dQ' = \overline{Q}_i dn_i + \overline{Q}_j dn_j + \overline{Q}_k dn_k + \cdots$$

- ✓ $\overline{Q}_i$ is the increase in the value of Q' for the mixture or solution when 1 mole of i is added to a large quantity of the solution at constant T and P.
- ✓ The stipulation that the quantity of solution be large is necessitated by the requirement that the addition of 1 mole of i to the solution should not cause a measurable change in its composition.

9.4. The Gibbs-Duhem Equation

✓ If $\overline{Q}_i$ is the value of Q per mole of i in the solution, then the value of Q' for the solution itself is

$$Q' = n_i\overline{Q}_i + n_j\overline{Q}_j + n_k\overline{Q}_k + \cdots$$

Differentiation

$$dQ' = \underline{n_i d\overline{Q}_i + n_j d\overline{Q}_j + n_k d\overline{Q}_k} + \cdots + \underline{\overline{Q}_i dn_i + \overline{Q}_j dn_j + \overline{Q}_k dn_k} + \cdots$$

$$dQ' = \underline{\overline{Q}_i dn_i + \overline{Q}_j dn_j + \overline{Q}_k dn_k} + \cdots$$

$$\underline{n_i d\overline{Q}_i + n_j d\overline{Q}_j + n_k d\overline{Q}_k} + \cdots = 0$$

9.4. The Gibbs-Duhem Equation

$$n_i d\overline{Q}_i + n_j d\overline{Q}_j + n_k d\overline{Q}_k + \cdots = 0$$

$$\sum n_i d\overline{Q}_i = 0$$

Division by *n*, the total number of moles of all the components of the solution

$$\sum_i X_i d\overline{Q}_i = 0$$

The generalized ***Gibbs–Duhem*** *equation.*

At constant temperature and pressure, the **partial molar Gibbs free energies** (**chemical potentials**) in a binary system are related as

$$X_A d\overline{G}_A + X_B d\overline{G}_B = 0$$

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9.5.1. The Molar and Partial Molar Gibbs Free Energy

$$Q' = n_i \overline{Q_i} + n_j \overline{Q_j} + n_k \overline{Q_k} + \cdots$$

✓ Applying to a **binary** solution, the Gibbs free energy at fixed temperature and pressure is

$$G' = n_A \overline{G}_A + n_B \overline{G}_B$$

$\overline{G_A}$ and $\overline{G_B}$ are the **partial** molar **G**ibbs free energies of A and B

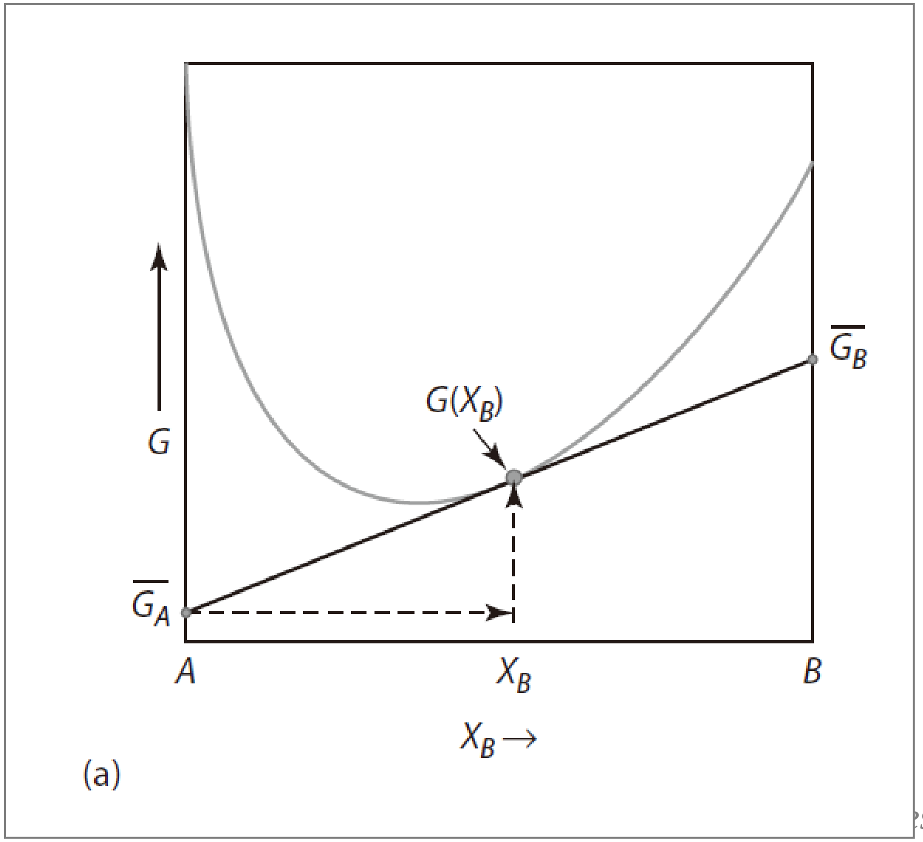
G' is the **total G**ibbs free energy of the solution

9.5.1. The Molar and Partial Molar Gibbs Free Energy

$$G' = n_A \overline{G}_A + n_B \overline{G}_B$$

Dividing both sides by (n_A+n_B) gives the molar Gibbs free energy of the solution as

$$G = X_A \overline{G}_A + X_B \overline{G}_B$$



$$G(X_B) = \overline{G}_A + X_B \frac{dG(X_B)}{dX_B}$$

$$\frac{dG(X_B)}{dX_B} = \overline{G}_B - \overline{G}_A$$

$$G(X_B) = \overline{G}_A + X_B\left(\overline{G}_B - \overline{G}_A\right)$$

9.5.1. The Molar and Partial Molar Gibbs Free Energy

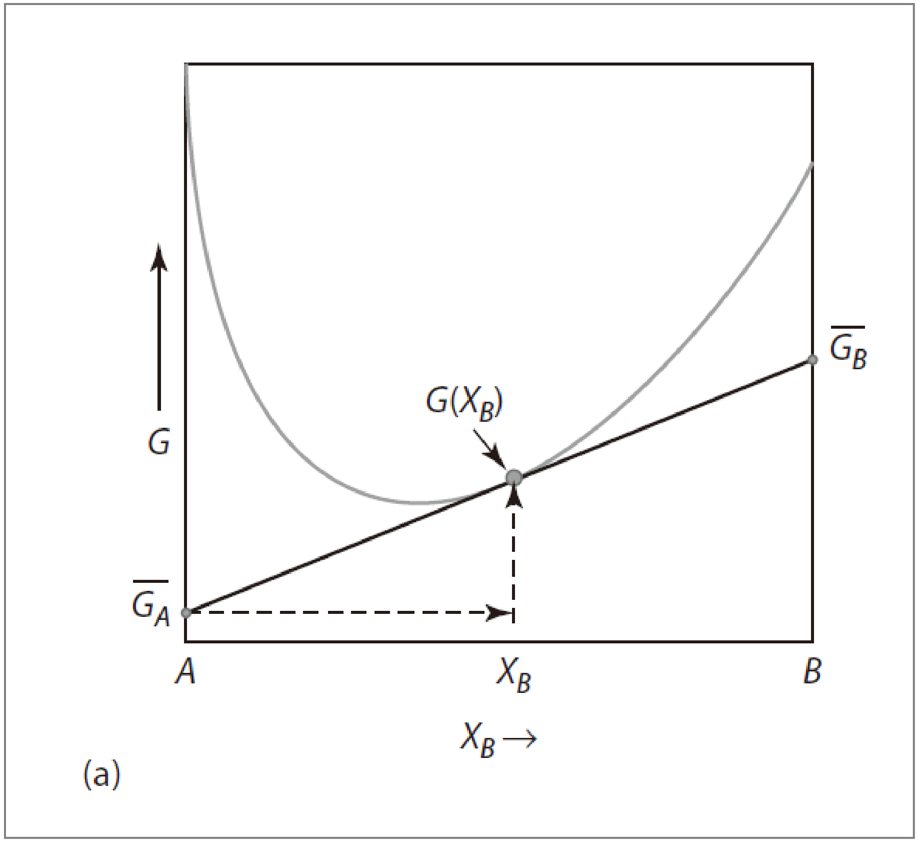


Figure 9.5 (a) the variation of the molar Gibbs free energy with composition, showing the partial molar Gibbs free energies of components A and B for the solution;

$$G(X_B) = X_A \overline{G}_A + X_B \overline{G}_B$$

$$\frac{dG(X_B)}{dX_B} = \overline{G}_B - \overline{G}_A$$

$$G(X_B) = \overline{G}_A + X_B(\overline{G}_B - \overline{G}_A)$$

$$\overline{G}_A = G - X_B \frac{dG}{dX_B}$$

$$\overline{G}_B = G + X_A \frac{dG}{dX_B}$$

9.5.1. The Molar and Partial Molar Gibbs Free Energy

$$\overline{G}_A = G - X_B \frac{dG}{dX_B}$$

$$\overline{G}_B = G + X_A \frac{dG}{dX_B}$$

- ✓ These expressions relate the dependence on composition of the **partial molar G**ibbs free energies of the components of a binary and the **molar G**ibbs free energy of the solution.

9.5.2. The Change in Gibbs Free energy due to the Formation of a Solution

- ✓ The pure component i, occurring in a condensed state at the temperature T, exerts an equilibrium vapor pressure, p_i^o and when occurring in a condensed solution at the temperature T, it exerts a lower equilibrium pressure, p_i.

- ✓ Consider the following **isothermal** three-step process:

(a). The **evaporation** of 1 mole of pure condensed i to vapor i at the pressure p_i^o

@const. T, P, equilibrium $\Delta G=0$ $\Delta G_{(a)}=0$

9.5.2. The Change in Gibbs Free energy due to the Formation of a Solution

(b). A **decrease** in the **p**ressure of 1 mole of vapor *i* from p_i^o to p_i

@const. T, P, equilibrium $\Delta G=0$ $\Delta G_{(b)}$ =?

$$G(P_2,T)-G(P_1,T)=RT\ln\frac{P_2}{P_1}$$

(c). The **condensation** of 1 mole of vapor *i* from the pressure p_i to the condensed solution

@const. T, P, equilibrium $\Delta G=0$ $\Delta G_{(c)}=0$

- ✓ The difference between the molar Gibbs free energy of *i* in the solution and the molar Gibbs free energy of pure *i* is given by the sum $\Delta G_{(a)} + \Delta G_{(b)} + \Delta G_{(c)}$.

9.5.2. The Change in Gibbs Free energy due to the Formation of a Solution

- ✓ Steps (a) and (c) are processes conducted at equilibrium, $\Delta G_{(a)}$ and $\Delta G_{(c)}$ are both = 0. The overall change in Gibbs free energy which accompanies the isothermal three-step process is thus $\Delta G_{(b)}$.

$$G(P_2,T)-G(P_1,T)=RT\ln\frac{P_2}{P_1} \quad \Rightarrow \quad \Delta G_{(b)}=RT\left(\ln\frac{P_i}{P_i^\circ}\right)$$

$$a_i=\frac{P_i}{P_i^\circ}$$

$$\Delta G(b)=G_i(\text{in solution})-G_i(\text{pure})=RT\ln a_i$$

$$\Delta G(b) = G_i(\text{in solution}) - G_i(\text{pure}) = RT\ln a_i$$

- ✓ But G_i (in solution) is simply the partial molar Gibbs free energy of i in the solution, and G_i (pure) is the molar Gibbs free energy of pure i.

- ✓ The **difference** between the two is the change in the Gibbs free energy accompanying the addition of 1 mole of i into the solution. This quantity is designated $\overline{\Delta G_i}^M$ and is the partial molar Gibbs free energy of **mixing** of the solution of i.

$$\Delta\overline{G}_i^M = \overline{G}_i - G_i^\circ = RT\ln a_i$$

9.5.2. the Change in Gibbs Free energy due to the Formation of a Solution

$$\Delta \overline{G}_i^M = \overline{G}_i - G_i^\circ = RT \ln a_i$$

- If n_A moles of *A* and n_B moles of *B* are mixed to form a solution at constant temperature and pressure,

The Gibbs free energy before mixing $= n_A G_A^\circ + n_B G_B^\circ$

The Gibbs free energy after mixing $= n_A \overline{G}_A + n_B \overline{G}_B$

- The **change** in the ***total*** Gibbs free energy caused by the mixing process, $\Delta G'^M$, sometimes referred to as the ***integral*** Gibbs free energy of mixing (total Gibbs free energy of mixing), **is the difference** between these quantities.

9.5.2. the Change in Gibbs Free energy due to the Formation of a Solution

$$\Delta G'^{M} = \left(n_A \overline{G}_A + n_B \overline{G}_B\right) - \left(n_A G_A^{\circ} + n_B G_B^{\circ}\right)$$

$$= n_A\left(\overline{G}_A - G_A^{\circ}\right) + n_B\left(\overline{G}_B - G_B^{\circ}\right)$$

$$\Delta G'^{M} = n_A \Delta\overline{G}_A^{M} + n_B \Delta\overline{G}_B^{M}$$

$$\Delta\overline{G}_i^{M} = RT \ln a_i$$

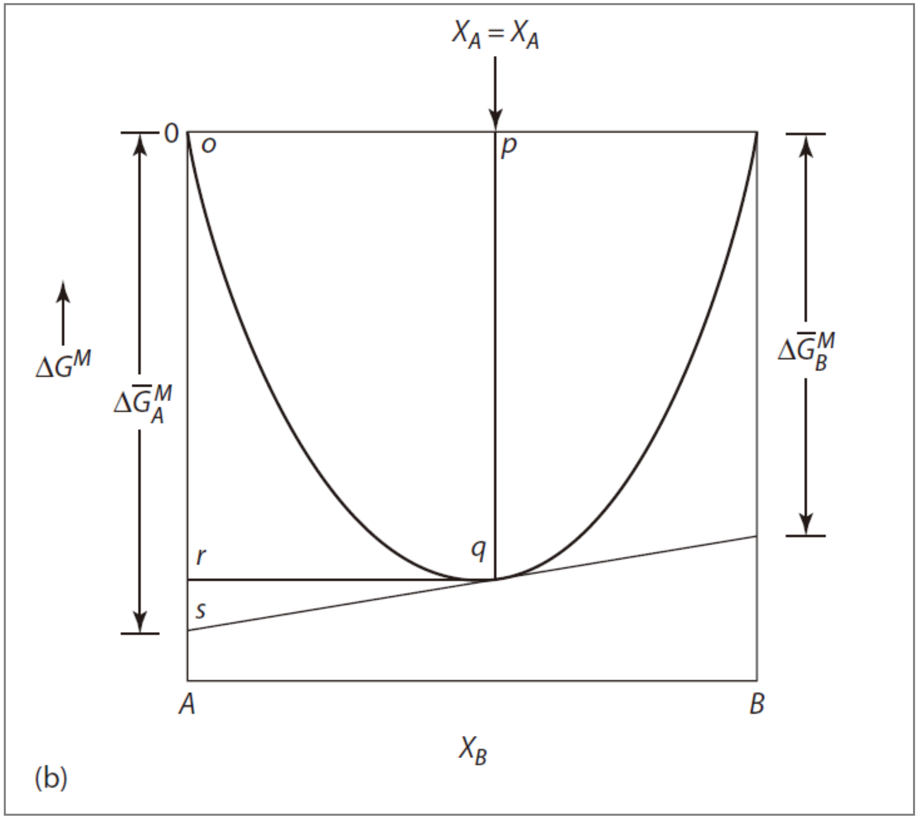
$$\Delta G'^{M} = RT\left(n_A \ln a_A + n_B \ln a_B\right)$$

9.5.2. the Change in Gibbs Free energy due to the Formation of a Solution

$$\Delta G'^{M} = n_A \Delta \overline{G}_A^M + n_B \Delta \overline{G}_B^M \Rightarrow \Delta G^M = X_A \Delta \overline{G}_A^M + X_B \Delta \overline{G}_B^M$$

For 1 mole of solution

$$\Delta G'^{M} = RT\left(n_A \ln a_A + n_B \ln a_B\right) \Rightarrow \Delta G^M = RT\left(X_A \ln a_A + X_B \ln a_B\right)$$



ΔG^M (X_B=q) is given by the segment $\overline{pq}$ and the segment $\overline{or}$.

Figure 9.5 (b) the variation, with composition, of the molar Gibbs free energy of formation (mixing) of a binary solution at a temperature, *T*.

The partial molar values for the alloy X_B=q are shown by the tangent intercepts with ordinate axes. At this temperature, there is complete solubility of A in B and B in A.

9.5.3 The Method of Tangential Intercepts

$$\overline{G}_A = G - X_B \frac{dG}{dX_B} \Rightarrow \Delta\overline{G}_A^M = \Delta G^M - X_B \frac{d\Delta G^M}{dX_B}$$

$$\overline{G}_B = G + X_A \frac{dG}{dX_B} \Rightarrow \Delta\overline{G}_B^M = \Delta G^M - X_A \frac{d\Delta G^M}{dX_A}$$

✓ Here, the partial molar Gibbs free energy of mixing of the components *A* and *B* can be read directly from the tangential intercepts at X_B=0 and X_B=1, respectively, just as the partial molar Gibbs free energies of X_B and X_A (their chemical potentials) can be read directly.

9.5. The Gibbs Free Energy of Formation of a Solution

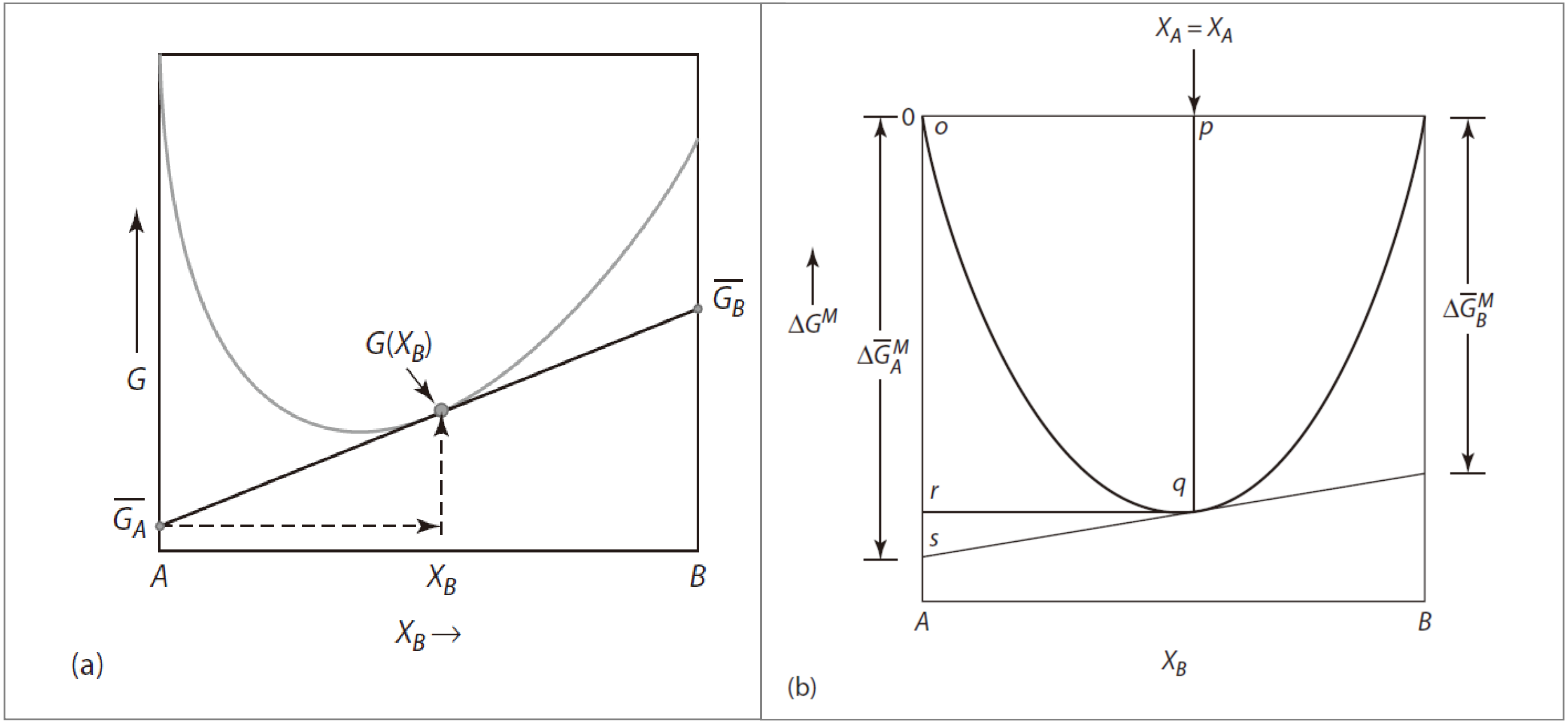


Figure 9.5a (a) The variation of the molar Gibbs free energy with composition, showing the partial molar Gibbs free energies of components *A* and *B* for the solution;(b) The variation, with composition, of the molar Gibbs free energy of formation(mixing) of a binary solution at a temperature, *T* . The partial molar values forthe alloy $X_B = q$ are shown by the tangent intercepts with ordinate axes. At this temperature, there is complete solubility of *A* in *B* and *B* in *A* .

Homework

1. When 1 mole of argon gas is bubbled through a large volume of an $F_e - M_n$ melt of $X_{Mn} = 0.5$ at 1863 K, the evaporation of Mn into the Ar causes the mass of the melt to decrease by 1.5 g. The gas leaves the melt at a pressure of 1 atm.Calculate the activity coefficient of Mn in the liquid alloy.

2. a. Calculate the values of $\Delta\bar{G}_B$ and a_B for an alloy of $X_B = 0.5$ at 1000 K assuming the solution is ideal .

b. Calculate the values of $\Delta\bar{G}_B$ and a_B for an alloy of $X_B = 0.5$ at 1000 K that is a regular solution with $\Delta H_{mixing} = 16628 X_A X_B$.

Homework

3. The variation, with composition, of G^{XS} for liquid Fe–Mn alloys at 1863 K is as follows

$\boldsymbol{X_{Mn}}$	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
$\boldsymbol{G^{XS}, J}$	395	703	925	1054	1100	1054	925	703	395

a. Does the system exhibit regular solution behavior?

b. Calculate $\bar{G}_{Fe}^{XS}$ and $\bar{G}_{Mn}^{XS}$ at $X_{Mn} = 0.6$.

c. Calculate ΔG^{M} at $X_{Mn} = 0.4$.

d. Calculate the partial pressures of Mn and Fe exerted by the alloy of $X_{Mn} = 0.2$.

The End